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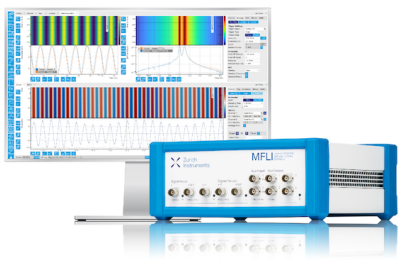
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The $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the isotopic lithium hydrides: The molecular constants, potential energy curves, and their adiabatic corrections

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The molecular constants and their adiabatic corrections have been determined for the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the isotopic lithium hydrides: ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD. Using a fully quantum mechanical variational method, the potential energy curves (IPA potentials) are determined. Extending the variational method, we have obtained for the first time adiabatic corrections of potential energy curves from isotopic spectroscopic data. A significant difference between the potential energy curves of the lithium hydrides and the lithium deuterides has been observed. When ^{6}Li was replaced by ^{7}Li, a significant difference was only observed for the $A\,^1\Sigma^+$ state, but not for the $X\,^1\Sigma^+$ state.

I. INTRODUCTION

The molecular constants and potential energy curves of the lithium hydrides have been of great interest to theoretical chemists and spectroscopists for quite some time since LiH represents the smallest neutral heteropolar molecule. In particular, the spectra of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system, which were first observed by Nakamura[1,2] and reinvestigated by Crawford and Jorgensen,[3,4] have attracted a lot of attention because of the anomalous behavior of the $A\,^1\Sigma^+$ state. As in all the other alkali metal hydrides, the ΔG_v and B_v values of the $A\,^1\Sigma^+$ state initially increase with increasing v (as manifested by a negative value of the $\omega_e x_e$ and α_e constants) before they eventually decrease in the usual manner. Since the early work of Mulliken,[5] it has been realized that this anomaly is due to an avoided crossing of the zero-order covalent and ionic potential energy curves. A more recent argument of Jenč,[6] who claimed that the anomaly is due to nonadiabatic effects, was disputed by Docken and Hinze[7] and does not agree with the results presented in this paper.

In addition, the lines of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system show an anomalous isotope shift,[8] which so far has prevented a unified description of the molecular constants and of the potential energy curves covering all the isotopic species. The reason for this is a breakdown of the Born–Oppenheimer approximation within the $A\,^1\Sigma^+$ state and, to a smaller extent, also within the $X\,^1\Sigma^+$ state,[9] as shown in this paper. As a consequence of this, the molecular constants of the different isotopes can no longer be described simply by using mass reduced variables[10] and the potential energy curves differ significantly from one isotope to the other.

It is the purpose of this paper to reanalyze the spectra of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the lithium hydrides and to present for the first time molecular constants and potential energy curves, including their adiabatic corrections which describe all the isotopic lithium hydrides which have been measured so far: ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD.

In the past, adiabatic corrections of potential energy curves[11,12] have been calculated theoretically or semitheoretically for several small molecules such as H_2^+,[13–22] H_2,[23,24] and HeH^+.[25,26] For some of the first and second row diatomic molecules (including the lithium hydrides), Wolfsberg and co-workers[27,28] have calculated corrections to the Born–Oppenheimer approximation for application to isotope exchange equilibria.

Adiabatic corrections of the Dunham coefficients have been determined directly from experiment for the ground state of CO[29] using optical heterodyne techniques. Similarly, adiabatic corrections have been obtained for the ground states of PbS,[30] GeS, and GeSe[31] from microwave measurements of different isotope combinations.

In the following section a set of Dunham-type coefficients and their adiabatic corrections are presented which for the first time fully describe the spectra of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of all the measured lithium hydrides. In the subsequent section, the corresponding potential energy curves are presented which are separated into a Born–Oppenheimer part and an adiabatic, mass dependent correction. For this purpose, an extension of a variational technique for determining potential energy curves[32] is used which is based on an inverted perturbation approach (IPA) first proposed by Kosman and Hinze.[33] It gives quantum mechanical potential energy curves which are not subject to the approximations generally associated with semiclassical potential energy curves. The latter ones are based on first and higher order WKB methods, such as the widely used Rydberg–Klein–Rees (RKR) technique. The paper presents, for the first time, adiabatic corrections of the potential energy curves based on spectroscopic measurements alone.

II. THE MOLECULAR CONSTANTS

In reanalyzing the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the lithium hydrides, we used the measurements of Crawford and

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Jorgensen,[3,4] Li and Stwalley,[9] and Orth and Stwalley[34] [see, also, the tables in K. C. Li and W. C. Stwalley, "Assigned Lines of the $A\ ^1\Sigma^+ - X\ ^1\Sigma^+$ Bands of ^{6}LiH, ^{6}LiD, ^{7}LiH, and ^{7}LiD," United States Energy Research and Development Administration Report COO-2326-20 (1976)]. In addition, microwave measurements of Pearson and Gordy[35] have been incorporated which improve, in particular, the rotational constants of the ground state. The aim of our analysis was to generate molecular constants which describe the measured spectra of the $A\ ^1\Sigma^+ - X\ ^1\Sigma^+$ system of all four observed isotope combinations.

In general, the line frequency $\bar{\nu}$ of a particular transition is given by

$$\bar{\nu} = T'(v',J') - T''(v'',J'') , \tag{1}$$

where $T'(v',J')$ and $T''(v'',J'')$ are the term values of the excited and the ground state, respectively. Within the Born–Oppenheimer approximation these term values can be expressed by[10]

$$T(v,J) = \sum_{ik} Y_{ik} \eta^i \xi^k , \tag{2}$$

where the "mass reduced quantum numbers"

$$\eta = (v+\tfrac{1}{2})\mu^{-1/2}$$

and

$$\xi = J(J+1)\mu^{-1} \tag{3}$$

have been introduced. Equations (1)–(3) describe the normal isotope shift as discussed, for example, by Herzberg.[36] This shift has frequently been used for the determination of the rotational and vibrational quantum numbers of molecular spectra. As soon as the Born–Oppenheimer approximation starts to break down, additional adiabatic correction terms have to be incorporated which give rise to an anomalous isotope shift. For a homonuclear diatomic molecule, the Dunham coefficients can then be expressed by means of an expansion in inverse powers of μ as follows[37]:

$$Y_{ik} = Y_{ik}^{(0)} + \mu^{-1} Y_{ik}^{(1)} + \mu^{-2} Y_{ik}^{(2)} + \cdots . \tag{4}$$

For a heteronuclear diatomic molecule like the lithium hydrides, the latter relation should be replaced by an expression of the form

$$Y_{ik} = Y_{ik}^{(0)} [1 + m_e(\Delta_{ik}^A/M_A + \Delta_{ik}^B/M_B) + \cdots] . \tag{5}$$

m_e is the electron mass and M_A and M_B the masses of the nuclei A and B, respectively. This was first pointed out by Watson[38] and has been overlooked in quite a number of papers, although it rarely had a noticeable effect in the analysis of measured spectra. To our knowledge, the only papers where significant adiabatic corrections to the Dunham coefficients have already been determined are those investigating the ground state of CO,[29,38] PbS,[30] GeS, and GeSe.[31] For Mg_2, only the electronic isotope shift, which modifies the spacing between the electronic states from one isotope combination to the other, came out to be statistically significant.[32] It gives rise to a small anomaly in the isotope shift which is manifested by a constant offset of the isotopic spectra.

TABLE I. The atomic masses in units of $M(^{12}C)/12$ according to Wapstra and Bos (Ref. 41), the reduced masses and the relative reduced masses $\mu_r = \mu/\mu(^7\text{LiH})$ for the lithium hydrides and lithium deuterides.

	H (1.007 825 037)	D (2.014 101 787)
^{6}Li	0.863 197 56	1.508 871 70
(6.015 123 2)	0.979 528 09	1.712 217 79
^{7}Li	0.881 238 19	1.564 870 53
(7.016 004 5)	1.0	1.775 763 41

The fact that Eq. (5) and not Eq. (4) is the correct relation to use for heteronuclear molecules is clearly demonstrated by Tiemann *et al.*[30] for the PbS molecule. In their Fig. 1, they have plotted μY_{01} as a function of μ^{-1} for all the measured isotope combinations. If Eq. (4) would have been correct, all their measurements should have been on a straight line. Instead, they obtain two sets of parallel lines depending on whether the Pb or S isotope is changed. This is only compatible with Eq. (5). Similar results will be presented below for the lithium hydrides.

In fitting the spectra of the $A\ ^1\Sigma^+ - X\ ^1\Sigma^+$ system we have described the term values according to Eqs. (2), (3), and (5) by

$$T(v,J) = \sum_{ik} \bar{A}_{ik} (v+\tfrac{1}{2})^i [J(J+1)]^k \mu_r^{-i/2-k} , \tag{6}$$

with

$$\bar{A}_{ik} = A_{ik} + B_{ik}^A/M_A + B_{ik}^B/M_B + \cdots , \tag{7}$$

where the A_{ik} are Dunham-type coefficients which are not necessarily identical with the mechanically defined Dunham coefficients Y_{ik}, as explained in detail by Hessel and Vidal.[39] Only if the analysis is restricted to term values close to the potential minimum where a perturbation approach becomes valid, do the A_{ik} coefficients go over into the Dunham coefficients Y_{ik}.[40] This difference is particularly significant in the $A\ ^1\Sigma^+$ state where the potential energy curve is extremely anharmonic.

The masses of the H, D, ^{6}Li, and ^{7}Li atoms in units of $M(^{12}C)/12$ have been taken from Wapstra and Bos.[41] They are given in Table I together with the reduced masses of the different isotope combinations and the relative reduced masses taken with respect to the most abundant ^{7}LiH molecule, $\mu_r = \mu/\mu\ (^7\text{LiH})$.

In our global fit of all data for determining the A_{ik} and B_{ik} in Eqs. (6) and (7) we had to consider the fact that data of different quality are used. In this case a weighted least squares fit has to be performed as explained by Albritton *et al.*[42] This was done in an iterative manner, where at the end of every fit the standard error of every subset of measurements σ_i is determined and used as a weighting factor for the subsequent least squares fit. Two to three iterations are generally enough to determine the standard errors within their own standard errors.[43]

The results of the weighted least squares fit are given in Table II for the $A\ ^1\Sigma^+$ state and in Table III for the

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TABLE II. The Dunham-type coefficients of the $A\,^1\Sigma^+$ state and their adiabatic corrections as obtained from a least squares fit of the measured line positions. All values are given in cm^{-1} together with their standard errors.

A'_{ik}	0	1	2	3	4	5	6
0	26 514.387 (0.18)	2.836 244 2 (0.001 6)	−1.407 162 04-3 (0.003 9-3)	1.439 813 53-6 (0.009 1-6)	−7.931 419-10 (0.10-10)	6.902 404-14 (0.61-14)	5.963 194-17 (0.21-17)
1	240.328 65 (0.32)	8.090 597-2 (0.28-2)	6.393 653 4-4 (0.071-4)	−1.517 887 8-6 (0.017-6)	1.240 842 2-9 (0.017-9)	−3.119 113 9-13 (0.061-13)	−1.904 657-17 (0.097-17)
2	26.491 082 (0.23)	−4.492 506-2 (0.20-2)	−1.853 872 9-4 (0.046-4)	6.439 639-7 (0.11-7)	−5.991 864-10 (0.11-10)	1.714 314 4-13 (0.038-13)	1.020 267-18 (0.11-18)
3	−3.754 623 9 (0.086)	1.615 563-2 (0.073-2)	2.513 219-5 (0.15-5)	−1.397 634 3-7 (0.030-7)	1.361 580 3-10 (0.029-10)	−3.796 950 3-14 (0.098-14)	
4	0.527 615 8 (0.019)	−3.674 848-3 (0.16-3)	−5.656 92-7 (2.8-7)	1.686 201-8 (0.046-8)	−1.650 751 6-11 (0.041-11)	4.139 889-15 (0.12-15)	
5	−5.977 297-2 (0.27-2)	5.098 047-4 (0.23-4)	−2.797 392-7 (0.36-7)	−1.136 593 4-9 (0.037-9)	1.085 505 2-12 (0.031-12)	−2.198 067 4-16 (0.076-16)	
6	4.605 512-3 (0.25-3)	−4.339 396-5 (0.21-5)	3.874 003-8 (0.32-8)	3.982 804-11 (0.16-11)	−3.601 978-14 (0.12-14)	4.527 170-18 (0.18-18)	
7	−2.219 923-4 (0.14-4)	2.205 030-6 (0.12-6)	−2.292 680-9 (0.18-9)	−5.616 977-13 (0.26-13)	4.626 066-16 (0.19-16)		
8	5.977 704-6 (0.45-6)	−6.110 492-8 (0.38-8)	6.562 414-11 (0.60-11)				
9	−6.812 360-8 (0.61-8)	7.060 497-10 (0.52-10)	−7.377 625-13 (0.83-13)				
B^H_{ik}	0	1	2				
0	2.179 668 (0.16)	−3.619 903-2 (0.14-2)	−6.773 40-6 (1.1-6)				
1	−6.127 295 (0.27)	4.392 772-2 (0.22-2)	2.163 291-6 (0.34-6)				
2	2.749 174 (0.17)	−2.216 906-2 (0.13-2)	−1.495 714-7 (0.23-7)				
3	−6.762 015-1 (0.51-1)	5.670 049-3 (0.38-3)					
4	9.434 593-2 (0.84-2)	−8.092 581-4 (0.61-4)					
5	−7.446 631-3 (0.78-3)	6.518 072-5 (0.55-5)					
6	3.090 393-4 (0.37-4)	−2.765 451-6 (0.26-6)					
7	−5.235 398-6 (0.71-6)	4.801 417-8 (0.49-8)					
B^{Li}_{ik}	0	1					
0	−1.630 (0.24)	−5.671-3 (0.72-3)					
1	−5.563-1 (0.29-1)						

$X\,^1\Sigma^+$ state. All A_{ik} and B_{ik} coefficients are given in cm^{-1} together with their standard errors in brackets. The larger number of excited state constants A'_{ik} and B'_{ik} immediately reflects the pronounced anomaly of the $A\,^1\Sigma^+$ state and is not merely a consequence of the larger data field employed.

We also note that for the molecular constants of the excited state we have to give a rather large number of digits. They are necessary to reproduce the measured line positions within the corresponding standard error. The reason for this is that, contrary to the molecular constants of the ground state, the higher order Dunham-type coefficients of the excited state are strongly correlated, indicating that the term values of the excited state are frequently dominated by small differences of the Dunham-type coefficients, which by themselves have a rather large standard error.

Table III contains also the total standard error and the rms errors for the different subsets of measurements. In particular, the rms error of the microwave measurements of 3.93×10^{-7} cm^{-1} is very much smaller than the error of the optical data and compares favorably with

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TABLE III. The Dunham-type coefficients of the $X\,^1\Sigma^+$ state and their adiabatic corrections as obtained from a least squares fit of the measured line positions. All values are given in cm^{-1} together with their standard errors.

A''_{ik}	0	1	2	3	4	5
0		7.519 496 6 (0.000 078)	−8.598 35-4 (0.003 0−4)	1.086 65-7 (0.0075-7)	−1.6544-11 (0.073-11)	1.7202-15 (0.021-15)
1	1405.910 93 (0.007 9)	−2.161 552-1 (0.000 14-1)	1.590 48-5 (0.007 7-5)	−9.350-10 (0.94-10)		
2	−23.090 81 (0.004 6)	1.776 3-3 (0.011-3)	3.48-8 (1.2-8)	−8.229-11 (0.89-11)		
3	0.144 23 (0.001 0)	2.901-5 (0.17-5)	−3.0543-8 (0.080-8)			
4	1.683-3 (0.10-3)	−5.3614-6 (0.076-6)				
5	−2.6091-4 (0.035-4)					

B^H_{ik}	0	1
0		−5.181-3 (0.15-3)
1	−4.7066-1 (0.039-1)	

B^{Li}_{ik}	0	1
0		−4.784-4 (0.16-4)

^{6}LiH: $N_{tot}=2238$, $\sigma_i=0.126\,23$
^{7}LiH: $N_{tot}=2303$, $\sigma_i=0.056\,27$
^{6}LiD: $N_{tot}=1710$, $\sigma_i=0.125\,09$
^{7}LiD: $N_{tot}=1151$, $\sigma_i=0.027\,91$
Microwave data: $N_{tot}=4$, $\sigma_i=3.93\times10^{-7}$
$N_{tot}=7406$, $\sigma=0.098\,31$

the error of 1.6×10^{-6} cm^{-1} estimated by Pearson and Gordy.[35] The noticeably smaller error of the least squares fit is most likely due to the fact that the least squares fit is rather insensitive to any possible systematic errors of the microwave measurements.[42]

It should also be noted that this kind of rms error could only be achieved if the least squares fit was based on Eqs. (6) and (7). When Eqs. (2), (3), and (4) were used, the rms error of the microwave measurements increased rather dramatically by three orders of magnitude. In accord with the earlier results of Ref. 30, this clearly indicates that Eqs. (6) and (7) and not Eqs. (2), (3), and (4) are the correct ones to use.

In Fig. 1 the data fields of the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ states are given for all four isotope combinations. They indicate the range of rotational-vibrational term values incorporated in the least squares fit. In case of the microwave measurements, only the first rotational transitions ($J''=0\rightarrow1$) of the lithium deuterides for $v''=0$ and 1 are available. Due to the small rms error, they

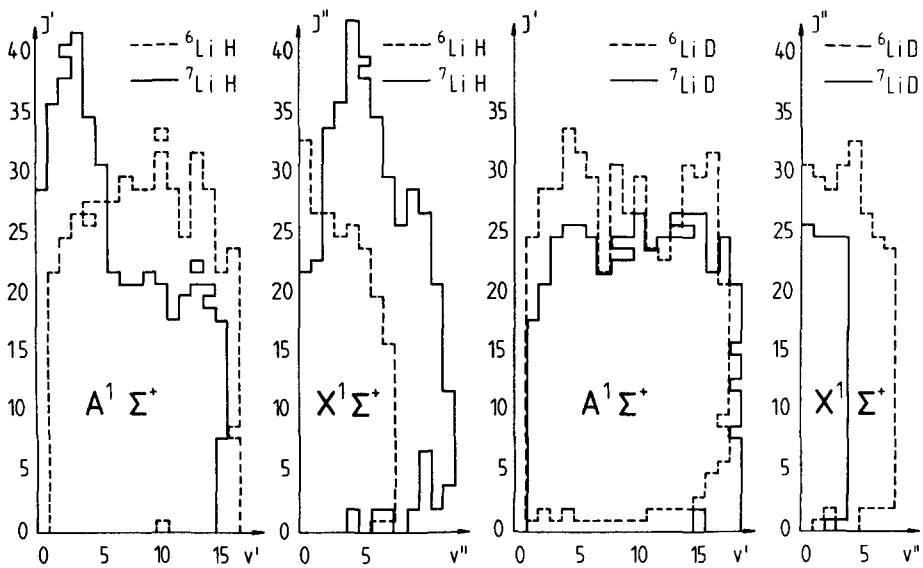


FIG. 1. Data fields of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the lithium hydrides: ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD. The levels indicated have been incorporated in the least squares fit.

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greatly improve the accuracy of the A''_{01} constant and, to a smaller extent, of the A''_{11} constant. As a result, the microwave data tend to stabilize the least squares fit as clearly demonstrated by the greatly reduced correlation of the A''_{01} and A''_{11} constants with the other constants determined in the fit.

In performing the global fit, 7406 lines have finally been adopted as shown in Table III. We eliminated all lines for which

$$\Delta\bar{\nu} = \bar{\nu}_{\text{meas}} - \bar{\nu}_{\text{calc}} > a\,\sigma_i \ .$$

σ_i is the rms error of the subset i and a was chosen to be 2.75. For a normal Gaussian distribution of the measurements, this should give a level of confidence of about 99%.[43] Instead, we had to eliminate from every subset of data between 6.3% and 9.3% of the data which amounted to a total of eliminated lines of 661. Since this number is rather large, in view of the expected level of confidence, we had to examine whether we inadvertently suppressed some information. A model based on Eqs. (6) and (7) tends to suppress, for example, possible perturbations.

In looking through the eliminated data, we found that the corresponding initial and final term values were *randomly* scattered over the entire data fields shown in Fig. 1. In addition, a sizable fraction of the eliminated lines were already marked as blends in the original measurements. The analysis also showed from the calculated overlap of some lines that some additional lines had apparently been overlooked as being blends. For the $A\ ^1\Sigma^+$ state only, a very few lines ended on levels which were slightly outside the data field of Fig. 1. For those term values outside the data field which were defined by more than one line, we were able to take combination differences since the lower state term values were all within the data fields of the $X\ ^1\Sigma^+$ state. It turned out that most of these combination differences were significantly larger than $3\sigma_i$ with σ_i being the rms error of the particular subset of measurements. This clearly indicates that at least some of these lines must have been incorrectly assigned or that they could not be measured accurately enough.

As a further test, we performed a separate least squares fit for every individual subset of data. It turned out that essentially the same lines had to be eliminated and that the total standard error of the individual fit was in excellent agreement with the rms error of that particular subset in the global fit. Furthermore, we performed a fit using the term value approach according to Aslund[44] in which the term values and not the Dunham-type coefficients are determined in a least squares fit. Such an approach is capable of handling perturbations. For all four isotope combinations we obtained the *same* standard error as in the case of the least squares fit which gives the Dunham type coefficients. The preceding evidence clearly indicates that the model based on Eqs. (6) and (7) did not eliminate any significant information.

Besides the A_{ik} constants, Tables II and III also show the adiabatic corrections of the Dunham-type coefficients B^{H}_{ik} and B^{Li}_{ik}, which are statistically significant.

TABLE IV. Summary of the most important molecular constants of the $A\ ^1\Sigma^+$ state for the lithium hydrides and lithium deuterides.

	^{6}Li H	^{7}Li H	^{6}Li D	^{7}Li D
$\bar{A}_{10}\,\mu_r^{1/2}$	234.16	234.17	237.19	237.21
$\bar{A}_{20}\,\mu_r$	29.22		27.86	
$\bar{A}_{30}\,\mu_r^{3/2}$	−4.426		−4.090	
$\bar{A}_{01}\,\mu_r$	2.7994	2.7995	2.8173	2.8175
$\bar{A}_{11}\,\mu_r^{3/2}$	0.1245		0.1027	
$\bar{A}_{21}\,\mu_r^2\times10^2$	−6.69		−5.59	
$\bar{A}_{02}\,\mu_r^2\times10^3$	−1.4139		−1.4105	
$\bar{A}_{10}$	236.59	234.17	181.27	178.01
$\bar{A}_{20}$	29.83	29.22	16.27	15.69
$\bar{A}_{30}$	−4.565	−4.426	−1.826	−1.729
$\bar{A}_{01}$	2.8579	2.7995	1.6454	1.5866
$\bar{A}_{11}$	0.1284	0.1245	0.0458	0.0434
$\bar{A}_{21}\times10^2$	−6.97	−6.69	−1.91	−1.77
$\bar{A}_{02}\times10^3$	−1.4736	−1.4139	−0.4811	−0.4473
$(\bar{A}_{02})_{\text{calc}}\times10^3$	−1.668	−1.600	−0.542	−0.504

As expected, they turn out to be quite a bit smaller for the $X\ ^1\Sigma^+$ state than for the $A\ ^1\Sigma^+$ state. As a result, the number of statistically significant B_{ik} constants is larger for the $A\ ^1\Sigma^+$ state than for the $X\ ^1\Sigma^+$ state. Furthermore, as one of the most important results we note that for both states

$$B^{\text{H}}_{ik} \gg B^{\text{Li}}_{ik} \ .$$

This shows that the adiabatic corrections are more sensitive to a change of the hydrogen isotopes than to a change of the lithium isotopes. This, however, is not simply a consequence of the larger change in the reduced mass of the system since the B_{ik} coefficients are clearly normalized with respect to the corresponding atomic mass, as shown in Eq. (7).

In order to further show the influence of the adiabatic corrections, Tables IV and V show some of the most important molecular constants for the $A\ ^1\Sigma^+$ and the $X\ ^1\Sigma^+$ state, respectively. They are obtained according to Eq. (7). These tables clearly indicate the origin of the anomalous isotope shift which is mainly due to the $A\ ^1\Sigma^+$ state. In addition, Tables IV and V contain $(A_{02})_{\text{calc}}$ as calculated by the Kratzer relation $D_e = 4B_e^3/\omega_e^2$. For the $A\ ^1\Sigma^+$ state, the agreement with the A_{02} constant as determined by the least squares fit is rather poor. This is certainly due to the anomalous behavior of the D_v terms which was pointed out by Li and Stwalley[8] (see their Fig. 8) although in the present analysis $D'_v(v'=0)$ does not differ as much from the higher $D'_v(v'>0)$ as in their analysis. For the $X\ ^1\Sigma^+$ state, however, the $(A_{02})_{\text{calc}}$ are in excellent agreement with the A_{02} constant of the least squares fit well within the standard error of the A_{02} constant.

It is interesting to note that for the $X\ ^1\Sigma^+$ state, the agreement between the A_{02} constants and the $(A_{02})_{\text{calc}}$ values is significantly better in the global fit than it is in the separate fit of the individual isotope combinations. This clearly shows that the additional mass dependence in the global fit, as incorporated in the mass-reduced variables and in the mass dependence associated

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TABLE V. Summary of the most important molecular constants of the $X\,^1\Sigma^+$ state for the lithium hydrides and lithium deuterides.

	^{6}LiH	^{7}LiH	^{6}LiD	^{7}LiD
$\bar{A}_{10}\,\mu_r^{1/2}$	1405.444		1405.677	
$\bar{A}_{20}\,\mu_r$	−23.091			
$\bar{A}_{30}\,\mu_r^{3/2}$	0.1442			
$\bar{A}_{01}\,\mu_r$	7.514 28	7.514 29	7.516 84	7.516 86
$\bar{A}_{11}\,\mu_r^{3/2}$	−0.216 155			
$\bar{A}_{21}\,\mu_r^2\times10^3$	1.776			
$\bar{A}_{02}\,\mu_r^2\times10^4$	−8.598			
$\bar{A}_{10}$	1420.055	1405.444	1074.252	1054.856
$\bar{A}_{20}$	−23.573	−23.091	−13.486	−13.003
$\bar{A}_{30}$	0.148 8	0.144 2	0.064 4	0.061 0
$\bar{A}_{01}$	7.671 32	7.514 29	4.390 12	4.233 03
$\bar{A}_{11}$	−0.222 967	−0.216 155	−0.096 478	−0.091 346
$\bar{A}_{21}\times10^3$	1.851	1.776	0.606	0.563
$\bar{A}_{02}\times10^4$	−8.962	−8.598	−2.933	−2.727
$(A_{02})_{\rm calc}\times10^4$	−8.955	−8.592	−2.933	−2.727

with the B_{ik} coefficients, tends to further constrain the physical model used in the fit. Hence, it improves the physical significance of the G_v and B_v values and of the centrifugal distortion terms D_v, H_v, ...

Similar to the results of Li and Stwalley,[8] we again obtained the anomalous behavior of the G_v' and B_v' values. For the $A\,^1\Sigma^+$ state of the lithium hydrides, ΔG_v shows the largest vibrational spacing between $v'=9$ and $v'=10$. For the ^{6}LiD molecule, the largest spacing occurs between $v'=12$ and $v'=13$ and for the ^{7}LiD molecule, between $v'=13$ and $v'=14$. For the lithium hydrides, B_v' goes through a maximum for $v'=3$ and for the lithium deuterides for $v'=4$ (see Tables X–XIII below).

III. THE POTENTIAL ENERGY CURVES

A. General

Since the lithium hydrides are the smallest neutral heteropolar molecules, a lot of work has gone into the determination of the potential energy curves. Among the numerous theoretical investigations, the calculations of Docken and Hinze[7] are presently the most widely accepted ones (see also the recent high quality results of Partridge and Langhoff[45]). Based on these calculations, Stwalley *et al.*[46] have constructed hybrid potential energy curves for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states. Around the potential minimum they used Rydberg-Klein–Rees (RKR) potentials. For the long range region, which goes to the accurately known dissociation limit, C_6, C_8, and C_{10} coefficients were used. The intermediate region was interpolated by scaling slightly the *ab initio* results of Docken and Hinze.[7]

Using RKR potential energy curves, Li and Stwalley[9] investigated the $X\,^1\Sigma^+$ state of the isotopic lithium hydrides. They found indications for the breakdown of the Born–Oppenheimer approximation which were manifested by small changes of the potential energy curves going from one isotope combination to the other. However, it was not clear whether the semiclassical approximation used in deriving the basic relations of the Rydberg–Klein–Rees method could have been responsible for these small differences between the potential energy curves of the different isotope combinations. In particular, one has to be careful since these differences in the potential energy curves are comparable to the differences between the measured term values and the quantum mechanical energy eigenvalues of the RKR potentials as shown below. In view of these difficulties, it is certainly desirable to reanalyze the potential energy curves of the lithium hydrides.

In trying to extract adiabatic corrections from RKR potentials, there is some ambiguity because the higher order corrections to the potential energy curves based on a first order WKB method[47] show a mass dependence similar to the adiabatic corrections. Hence, it is difficult to disentangle the adiabatic corrections from the higher order corrections to the RKR potentials. For this reason, the potential energy curves are reanalyzed in the following, using a variational method recently developed by Vidal and Scheingraber[32] which is based on an inverted perturbation approach (IPA) first proposed by Kosman and Hinze.[33] These potential energy curves are fully quantum mechanical curves and do not suffer from the mass dependent correction terms of the first and higher order WKB methods.[47]

There is a further reason for ambiguities in the analysis of RKR potentials. The RKR potentials are generally based on G_v and B_v values which are determined in a least squares fit of the spectroscopic data. In such a fit, the G_v and B_v values depend to some extent on the number of centrifugal distortion terms used in the fit and hence, depend on the data field which has been accepted in the least squares fit. Since the G_v and B_v values originate from an extrapolation to $J=0$, investigators have frequently tried to improve the physical significance of the G_v and B_v values by suppressing measurements which involve transitions between levels with elevated values of J. This is again another drawback of the RKR potentials which is avoided in the variational method for determining the quantum mechanical potential energy curves.[32] In the latter method, the Dunham-type coefficients are only used for reproducing the measured term values and they are not used for an extrapolation to $J=0$ which, in most cases, is outside the measured data field.

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B. Theory

In solving the Schrödinger equation of a diatomic molecule, one generally makes the adiabatic approximation[11,48]

$$\psi(r,R)=\phi(r;R)\,\psi_{vJ}(R)\ , \tag{8}$$

in which the electronic and nuclear motion of the molecule are separated. R describes the internuclear separation and r refers to the coordinates of the electrons. If one neglects the effect of the kinetic energy operator $\partial^2/\partial R^2$ on the electronic part of the wave function $\phi(r;R)$, the Schrödinger equation is separable and one of the two resulting equations gives rise to the linear Schrödinger equation of a rotating vibrator

$$(H_0+H_{\rm rot})\,\psi_{vJ}(R)=E_{vJ}\psi_{vJ}(R)\ . \tag{9}$$

H_0 is the Hamiltonian of the nonrotating molecule

$$H_0=-\frac{\hbar}{4\pi\mu c}\frac{\partial^2}{\partial R^2}+U_0(R) \tag{10}$$

and $U_0(R)$ is the potential of the rotationless molecule. The energy is given in cm^{-1}. Since we are dealing with $^1\Sigma$ states, we have neglected in Eq. (9) a term responsible for the fine structure. $H_{\rm rot}$ describes the rotational motion of the nuclei and for a $^1\Sigma$ state we have

$$H_{\rm rot}=\frac{\hbar}{4\pi\mu c}\frac{J(J+1)}{R^2}\ . \tag{11}$$

It is now important to note that the concept of an effective molecular potential $U_0(R)$ is only valid within the adiabatic approximation where nonadiabatic effects, which are due to interactions between different electronic states, can be neglected. Within the adiabatic approximation, $U_0(R)$ is given by[11]

$$U_0(R)=U_{\rm BO}(R)+(M_A^{-1}+M_B^{-1})\,\Delta U^{+}(R)+(M_A^{-1}-M_B^{-1})\,\Delta U^{-}(R)+\cdots\ . \tag{12}$$

Within the Born–Oppenheimer approximation or "fixed nuclei" approximation, where the nuclei become infinitely heavy, $U_0(R)$ simplifies even further and goes over into the Born–Oppenheimer potential $U_{\rm BO}(R)$, which is identical for all different isotope combinations of a particular molecule. $U_{\rm BO}(R)$ gives rise to the normal isotope shift as defined by Eqs. (1)–(3). To the extent that the adiabatic corrections $\Delta U^{+}(R)$ and $\Delta U^{-}(R)$ become important, the effective potential energy curve $U_0(R)$ differs from one isotope combination to the other and the isotope shift becomes anomalous. For a homonuclear diatomic molecule $\Delta U^{-}(R)$ in Eq. (12) vanishes, of course. Equation (12) can also be written as

$$U_0(R)=U_{\rm BO}(R)+M_A^{-1}\Delta U^{A}(R)+M_B^{-1}\Delta U^{B}(R)+\cdots\ , \tag{13}$$

where

$$\Delta U^{A}(R)=\Delta U^{+}(R)+\Delta U^{-}(R)\ ,\qquad \Delta U^{B}(R)=\Delta U^{+}(R)-\Delta U^{-}(R)\ . \tag{14}$$

Theoretical calculations generally have employed Eq. (12). In our case of the lithium hydrides, however, it turns out that the standard error of $\Delta U^{\rm Li}(R)$ is much larger than the standard error of $\Delta U^{\rm H}(R)$. This is directly related to the fact that the change of the reduced mass is smaller interchanging the lithium isotopes than interchanging the hydrogen isotopes. For this reason, we have carried out our analysis of the potential energy curves on the basis of Eq. (13) as shown below.

C. Analysis of the individual isotope combinations

Before we try to evaluate the Born–Oppenheimer potential $U_{\rm BO}(R)$ and the adiabatic corrections $\Delta U(R)$, we first analyze the effective potentials $U_0(R)$ for the electronic states of the different individual isotope combinations separately. The purpose of this is to demonstrate, first of all, to what extent the adiabatic approximation and, consequently, the concept of an effective potential, is valid. This was done using the variational method where $U_0(R)$ in Eq. (9) is varied in such a manner that the quantum mechanical energy eigenvalues E_{vJ} agree with the measured term values $T(v,J)$ in a least squares sense. For this purpose, we start with some initial potential $U_i(R)$ which is generally taken to be an RKR potential. For this initial potential, a correction $\Delta U_0(R)$ is determined such that an improved potential $U_f(R)$ is obtained, where

$$U_f(R)=U_i(R)+\Delta U_0(R)\ . \tag{15}$$

According to first order perturbation theory, the potential energy correction $\Delta U_0(R)$ is related to the energy level correction $\Delta E_{vJ}=T(v,J)-E^0_{vJ}$ by[33]

$$\Delta E_{vJ}=\langle\psi^0_{vJ}\,|\,\Delta U_0(R)\,|\,\psi^0_{vJ}\rangle\ , \tag{16}$$

where ψ^0_{vJ} is the radial wave function and E^0_{vJ} the energy eigenvalue of the Schrödinger Eq. (9) with $U_0(R)=U_i(R)$. In order to obtain a correction function $\Delta U_0(R)$ which is sufficiently flexible, it is generally most convenient to describe $\Delta U_0(R)$ as a linear combination of some functions $f_i(R)$,

$$\Delta U_0(R)=\sum_i C_i f_i(R)\ . \tag{17}$$

In this case, the coefficients C_i are given by a system of linear equations

$$\Delta E_{vJ}=T(v,J)-E^0_{vJ}=\sum_i C_i\,\langle\psi^0_{vJ}\,|\,f_i(R)\,|\,\psi^0_{vJ}\rangle\ . \tag{18}$$

The coefficients are generally highly overdetermined and can be obtained by a least squares method. For $f_i(R)$, we used Legendre polynomials with a suitable exponential cutoff and a nonlinear transformation of the internuclear distance, as described in Ref. 32.

For the initial potential energy curves of the $A\ ^1\Sigma^+$ and $X\ ^1\Sigma^+$ state, we used the RKR potentials based on the G_v and B_v values which are calculated from the $\bar{A}_{i0}$ and $\bar{A}_{i1}$ coefficients of the different isotopes. For all these RKR potentials, we determined the rms error

$$\sigma_E=\sqrt{\sum_n(\Delta E_n)^2/n}\ , \tag{19}$$

where n is the number of term values used in the variational method and ΔE_n is the difference between the measured term value $T(v,J)$ and the quantum mechanical energy eigenvalue E^0_{vJ}. It turned out that for the $X\ ^1\Sigma^+$ state of the four different isotope combinations, σ_E(RKR) ranged from 0.02–0.20 cm^{-1}, whereas for the $A\ ^1\Sigma^+$

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TABLE VI. List of J values used in the variational method for generating the IPA potentials. The J values are followed in brackets by the range of vibrational quantum numbers employed.

$A\,^1\Sigma^+$ state	
^{6}Li H:	2,6,10,14,17(1–16), 20(1–15), 24(1–14), 28(1–10).
^{7}Li H:	2(1–15), 6,11,16(0–15), 20(1–15), 24(1–14), 28,30, 34(1–4), 40(2–3).
^{6}Li D:	2,6,10,14,17,20(1–17), 24,28(2–16), 30(3–15).
^{7}Li D:	2,5,8,11,14,17(1–18), 20(2–18), 24(2–17).
$X\,^1\Sigma^+$ state	
^{6}Li H:	2,6,10,14,17(0–7), 20(0–6), 23,26(0–5), 28(0–1), 30(0).
^{7}Li H:	2,6,10(0–12), 14,17,20(0–11), 23(0–10), 25(0–9), 32(0–6), 40(2–3).
^{6}Li D:	2,6,10,14,17,20,23(0–7), 26(0–5), 29(0–4).
^{7}Li D:	2,6,10,14,17,20,23,24(0–3).

state, the rms error ranged from 5.9–11.5 cm^{-1}. The latter values are surprisingly large and comparable to the adiabatic corrections to be determined below. While the A'_{00} coefficients (see below) account for part of this $A\,^1\Sigma^+$ rms error, the remaining error is still quite sizable. It immediately shows that adiabatic corrections based on these RKR potentials are highly questionable.

After applying the variational method where we used typically between 9 and 16 different $f_i(R)$ functions, we obtained a rather dramatic improvement. For the $X\,^1\Sigma^+$ state the rms error σ_E(IPA) was in the range of 0.01–0.03 cm^{-1} and for the $A\,^1\Sigma^+$ state it was in the range of 0.01–0.04 cm^{-1}. For the $A\,^1\Sigma^+$ state, this is an improvement by more than two orders of magnitude. In all cases, the rms error σ_E(IPA) was well within the corresponding rms error of the global least squares fit σ_i.

The data fields used in the variational method are summarized in Table VI, which gives for every isotope combination the quantum numbers J which are followed in brackets by the range of vibrational quantum numbers employed. Table VII contains under the heading "analysis of individual isotope combinations" the rms error of the RKR potential σ_E(RKR) and the rms error obtained for the IPA potential σ_E(IPA). All values are given in cm^{-1} for the $A\,^1\Sigma^+$ as well as the $X\,^1\Sigma^+$ state together with the corresponding rms errors of the global least squares fit and of the fit using the term value approach.

It should be pointed out that the σ_E(IPA) being smaller than σ_i has no special significance. This is simply an artifact of the Dunham-type coefficients. In fitting the measured line positions $\bar{\nu}$ by the A_{ik} and B_{ik} coefficients, a lot of the noise of the measurements has been eliminated and it is not surprising that a fit of the quantum mechanical energy eigenvalues to term values, which are defined by Eqs. (6) and (7), produces a σ_E which is smaller than σ_i. This is particularly obvious for electronic states with a rather small data field like, for example, the $X\,^1\Sigma^+$ state of the ^{7}LiD molecule.

Therefore, it would be more appropriate to circumvent the intermediate step of the Dunham-type coefficients altogether and to fit the measured line positions directly to the difference of the term values of two potential energy curves, which have to be adjusted simultaneously. This requires, however, rather extensive calculations because the Schrödinger equation has to be solved for every level within the data field of the measurements. In this case, one obtains $\sigma_E = \sigma_i$ if the model used in the least squares fit and the Hamiltonian are both adequate. Work on this is in progress.[49]

To further illustrate the improvement which has been achieved using the variational method, we have shown in Fig. 2 for the $A\,^1\Sigma^+$ state of the ^{7}LiH molecule the energy difference ΔE_{vJ} as a function of the vibrational quantum number v for the different J used in the variational method (see Table VI). ΔE_{vJ} is defined to be the difference between the quantum mechanical energy eigenvalue E_{vJ} and the measured term value $T(v,J)$ as specified by the Dunham-type coefficients A_{ik} and B_{ik} in Tables II and III. The results are very similar for the $A\,^1\Sigma^+$ state of the other isotope combinations. In Fig. 2 we have only shown the energy differences ΔE_{vJ} for the RKR potential because the final IPA potential gives rise to energy differences which are smaller than 0.1 cm^{-1} and which would hardly show up in the linear plot of Fig. 2.

It is also interesting to note in Table VII that in the $A\,^1\Sigma^+$ state the rms error for the RKR potentials σ_E(RKR) is about twice as large for the lithium hydrides as it is for the lithium deuterides. This supports the

TABLE VII. Summary of the rms errors of the energy differences ΔE_{vJ} for the RKR and IPA potentials for the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ states.

Molecule	^{6}Li H	^{7}Li H	^{6}Li D	^{7}Li D
σ_i (least squares fit)	0.1262	0.0563	0.1251	0.0279
σ_i (term value approach)	0.1225	0.0519	0.1184	0.0276
	$A\,^1\Sigma^+$ state			
	Analysis of individual isotope combinations			
σ_E(RKR)	11.547	11.524	6.069	5.867
σ_E(IPA)	0.0387	0.0426	0.0254	0.0142
	Global analysis			
σ_E(RKR)	10.800	10.874	11.380	11.331
σ_E(IPA)	0.0360	0.0291	0.0295	0.0126
σ_E(total)	0.0286			
	$X\,^1\Sigma^+$ state			
	Analysis of individual isotope combinations			
σ_E(RKR)	0.105	0.203	0.067	0.018
σ_E(IPA)	0.0142	0.0329	0.0148	0.0031
	Global Analysis			
σ_E(RKR)	0.107	0.192	1.236	0.793
σ_E(IPA)	0.0138	0.0359	0.0222	0.0086
σ_E(total)	0.0270			

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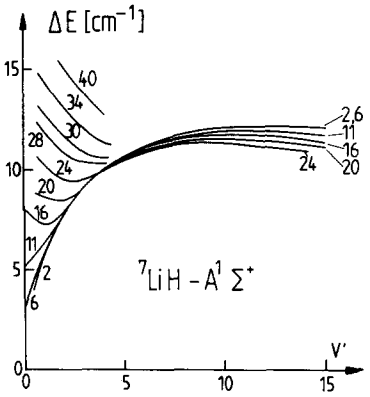


FIG. 2. The energy difference between the quantum mechanical energy eigenvalues of the RKR potential and the measured term values as a function of the vibrational quantum numbers v for different J values. The results are given for the ^{7}LiH $A\,^1\Sigma^+$ state.

obvious conclusion that a semiclassical first order WKB method, which forms the basis for the RKR potentials, holds better for the heavier molecules than for the lighter ones.

The most important conclusion which can be drawn from Table VII, however, is the fact that in all cases one can provide an effective potential energy curve whose quantum mechanical energy eigenvalues are capable of describing the measured term values within 3 σ_i over the *entire* data field, where σ_i is the rms error of the subset i in the least squares fit. As discussed above it is σ_i and not σ_E which is the relevant quantity for judging the quality of the fit.

The fact that within the accuracy of the measurements, *all* measurements can be described by an effective potential energy curve is by no means trivial if the potential energy curves of the lithium hydrides are due to an avoided crossing of the zero-order covalent and ionic potential energy curves as first proposed by Mulliken[5] and recently discussed again, for example, by Herschbach and co-workers.[50,51]

It should be noted that the concept of an effective potential is partially capable of absorbing some nonadiabatic effects. This means that within the accuracy of the measurements an adiabatic approximation can be carried out over the entire data field. This implies that the electronic part of the wave function $\phi(r;R)$ in Eq. (8), which depends parametrically on the internuclear separation R, changes sufficiently slowly as a function of R such that the effects of the kinetic energy operator $\partial^2/\partial R^2$ on $\phi(r;R)$ can be neglected. This demonstrates that the electronic part of the wave function may well change its character from covalent to ionic or vice versa, as long as this happens slowly enough.

D. Global analysis

After verifying that effective potential energy curves do exist for the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ states of all four isotope combinations, we then tried to extract the adiabatic corrections by applying Eqs. (12) or (13) to the potential energy curves of the individual isotope combinations. However, it turned out that this method failed and gave unphysical results. The problem is associated with the accuracy of the potential energy curves. For a given potential $U_i(R)$ in Eq. (15), the standard error of the final potential $U_f(R)$ is given by the standard errors and the correlation matrix of the C_i coefficients in Eq. (17) as shown by Vidal[52] (see also Ref. 49). In case of the $A\,^1\Sigma^+$ state the standard error around the innermost turning points at 1.41 Å was, for example, 7 cm^{-1} for a standard error σ_E(IPA) = 0.03 cm^{-1}. This implies that the potential energy curve at $R = 1.41$ Å can be changed by as much as 7 cm^{-1} without affecting the energy eigenvalues by more than 0.03 cm^{-1}. This is simply a consequence of the steepness of the potential energy curve at these inner turning points.

Since the adiabatic corrections are comparable to these standard errors as shown below, it is obvious why a direct fit of Eqs. (12) or (13) to the potential energy curves of the individual isotope combinations can be rather ambiguous.

For this reason, we then proceeded in a different manner. Similar to the least squares fit of the previous section, we performed a global analysis of all four different isotope combinations by inserting Eqs. (12) or (13) into the Hamiltonian of the rotationless molecule H_0 as specified by Eq. (10). In the subsequent variational method, we simultaneously adjusted $U_{BO}(R)$ and the two different adiabatic corrections $\Delta U^H(R)$ and $\Delta U^{Li}(R)$.

It should be noted at the outset that the problem of determining the potential energy curves according to Eqs. (12) or (13) is overdetermined due to the knowledge of four instead of three different isotope combinations. On the other hand, this redundancy was removed at the intermediate step of determining the Dunham-type coefficients because the spectra of all four isotope combinations could be described by the A_{ik}, B^H_{ik}, and the B^{Li}_{ik} coefficients. No further coefficients were required which were statistically significant. Since in our global analysis of the potential energy curves the measured term values are defined through the A_{ik} and B_{ik} coefficients only, the problem is now no longer overdetermined.

In carrying out the variational procedure, we used as an initial potential for $U_{BO}(R)$ the RKR potential of the ^{7}LiH molecule and for the initial adiabatic corrections we have set $\Delta U(R) = 0$. In this manner, the variational method converged just as rapidly as for the individual isotope combinations. Three iterations were required. In the case of the $A\,^1\Sigma^+$ state we used 21 C_i coefficients for $U_{BO}(R)$, seven C_i coefficients for $\Delta U^H(R)$ and two C_i coefficients for $\Delta U^{Li}(R)$; and in the case of the $X\,^1\Sigma^+$ state we used ten C_i coefficients for $U_{BO}(R)$, six C_i coefficients for $\Delta U^H(R)$ and one C_i coefficient for $\Delta U^{Li}(R)$. Furthermore, the same values of J and v which are listed in Table VI were used as for the analysis of the individual isotope combinations.

Similar to our global least squares fit, we also had to

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account for the fact that we are employing measurements of different inherent accuracy. In solving the overdetermined system of linear equations for the different C_i coefficients according to Eq. (18), we performed again a weighted least squares fit. The rms error for a particular subset of data, which is associated with an individual isotope combination, is used as a weighting factor for the subsequent analysis. This method converged typically after two to three iterations.

The results of our global analysis are given in Table VII under the heading "global analysis," where they can be compared with the previous results of the individual fits discussed above.

As a first remarkable result, we note that for the $X\,^1\Sigma^+$ state the adiabatic correction due to the lithium isotopes is not statistically significant except for a constant C_1 [$f_1(R)=1$], which will be discussed below. This means that within the accuracy of the measurements, the shape of the $X\,^1\Sigma^+$ potential energy curve is independent of the particular lithium isotope. For the $A\,^1\Sigma^+$ state, however, we obtained a statistically significant adiabatic correction due to the lithium isotopes.

Furthermore, the adiabatic correction due to the hydrogen atom is statistically significant for both electronic states. For this reason, we have carried out our final analysis on the basis of Eq. (13) which appears to be more appropriate than Eq. (12) since for the $X\,^1\Sigma^+$ state with $\Delta U^{\mathrm{Li}}(R)=0$ we have $\Delta U^{+}(R)=\Delta U^{-}(R)$ within the standard errors. These results correspond to the results of the previous section where we noted that $B^{\mathrm{H}}_{ik} \gg B^{\mathrm{Li}}_{ik}$. We also note that in our global analysis of an electronic state the different rms errors σ_E(IPA) are essentially identical with those which we obtained in the analysis of the individual isotope combinations.

At this point, it is interesting to come back to Fig. 2. There it is immediately apparent that the rms error σ_E(RKR) could have been reduced by changing the $\bar{A}_{00}$ constant. For the $\bar{A}_{00}$ constant we took the Y_{00} constant as defined by Dunham

$$Y_{00}=\frac{B_e-\omega_e x_e}{4}+\frac{\alpha_e\omega_e}{12B_e}+\frac{\alpha_e^2\cdot\omega_e^2}{144B_e^3}\,. \tag{20}$$

After going through the variational procedure, one can give improved A_{00} coefficients which define the measured term values with respect to the potential minimum. Simultaneously, one obtains the electronic term difference T_e which is the difference between the energies of the corresponding potential minima. This quantity cannot be determined directly in a least squares fit. The A^*_{00} constant in Table II contains A'_{00}, A''_{00}, and T_e according to

$$A^*_{00}=A'_{00}+T_e-A''_{00}\,. \tag{21}$$

As another interesting aspect, it should be noted that in the variational method the shape of the potential energy curve around the minimum is the most accurately determined part because it has the strongest influence on the quantum mechanical energy eigenvalues. As a result of this, the standard error of the molecular potential as defined by the standard errors and the correlation matrix of the C_i coefficients goes through a minimum around the potential minimum. This is in contrast to the RKR method, where the potential minimum is the least accurately known part of the molecular potential and originates essentially from an extrapolation which generally gives good results for "well-behaved" potentials with a sufficiently small anharmonicity where a perturbation approach can be successfully applied.

TABLE VIII. Summary of the Dunham coefficients Y_{00} and of the A_{00} coefficients as obtained from the variational method. Based on the A_{00} coefficients, the electronic term difference T_e, and the zero point energy E_0 is given.

Molecule	^{6}Li H	^{7}Li H	^{6}Li D	^{7}Li D
A^*_{00}	26 516.28	26 516.32	26 515.20	26 515.24
		$A\,^1\Sigma^+$ state		
Y'_{00}	7.561	7.406	4.165	4.016
A'_{00}	7.608	7.544	3.322	3.263
E'_0	132.73	131.26	98.22	96.33
		$X\,^1\Sigma^+$ state		
Y''_{00}	1.006	0.985	0.575	0.554
A''_{00}	1.010	0.990	0.566	0.547
E''_0	705.17	697.96	534.33	524.74
T_e	26 509.68	26 509.77	26 512.44	26 512.52

In Table VIII, we have summarized for the $A\,^1\Sigma^+$ and the $X\,^1\Sigma^+$ state the values of the A'_{00}, A''_{00}, and T_e constants which are related according to Eq. (21) together with the Y'_{00} and Y''_{00} constants according to Eq. (20). In addition, we have included the energies of the lowest vibrational level with respect to the potential minimum. Similar results can, of course, also be extracted from the above analysis of the individual isotope combinations. However, in view of the smaller degree of freedom in our global analysis, we report the latter values which differ only very little within the standard errors from those obtained in the analysis of the individual isotope combinations.

For both electronic states, we notice a difference between the A'_{00} and the Y'_{00} constants of less than 4%. In the case of the electronic term difference T_e which gives the energy difference between the potential minima of the $A\,^1\Sigma^+$ and the $X\,^1\Sigma^+$ state, we obtain a change of 2.75 cm^{-1} replacing the hydrogen atom by the deuterium atom, whereas we obtain a change of 0.08 cm^{-1} replacing the lithium isotopes. The latter change is smaller than the standard error of A^*_{00} in Table II and is, therefore, not statistically significant. In going from the hydrides to the deuterides, on the other hand, we observe a significant change in the shape of the potential energy curves as well as a change in the electronic term difference T_e.

In performing our global analysis of all isotope combinations, we noted that the only significant C_i coefficients of the adiabatic corrections due to the lithium isotopes was the C_1 constant for the $X\,^1\Sigma^+$ state and the C_1 and C_2 constants for the $A\,^1\Sigma^+$ state. C_1 is the coefficient of the Legendre polynomial $P_1(R)=1$. This

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TABLE IX. Dissociation energies (in cm^{-1}) for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of the lithium hydrides and deuterides.

Isotope	D_0 (cm^{-1})		D_e (cm^{-1})	
	$X\,^1\Sigma^+$	$A\,^1\Sigma^+$	$X\,^1\Sigma^+$	$A\,^1\Sigma^+$
^{6}Li H[a]	19581.6 ± 0.6	8548.1 ± 0.6	(20286.8 ± 0.6)[a]	(8680.8 ± 0.6)[a]
^{7}Li H	19588.8 ± 0.5	8549.4 ± 0.5	20286.8 ± 0.5	8680.7 ± 0.5
^{6}Li D[a]	19758.3 ± 0.5	8585.6 ± 0.5	(20292.6 ± 0.5)[a]	(8683.8 ± 0.5)[a]
^{7}Li D	19767.8 ± 0.4	8587.4 ± 0.4	20292.6 ± 0.4	8683.7 ± 0.4

[a]Assuming $D_e(^6\text{Li H}) = D_e(^7\text{Li H}) \pm 0.1\ cm^{-1}$, $D_e(^6\text{Li D}) = D_e(^7\text{Li D}) \pm 0.1\ cm^{-1}$.

term gives rise to a constant shift of the whole potential energy curve. It has, however, no physical significance because the constant shift simply depends on an arbitrary definition of how one defines the energy of one isotope combination with respect to the other ones.

In theoretical work, it is generally convenient to define the zero point of the common energy scale as the limit where the nuclei and the electrons of the molecule are all infinitely far apart. Applying this convention requires a knowledge of the atomic ionization energies and of the dissociation energies. The dissociation energies of the molecular ground state are rather well known from an analysis of the shallow $B\,^1\Pi$ state for the ^{7}LiH[53] and for the ^{7}LiD molecule.[54] They are not accurately enough known for the isotope combinations containing the ^{8}Li isotope although to a good approximation it appears justified to adopt for the ^{6}Li isotope the values obtained for the isotope combinations containing the ^{7}Li isotope. Since, on the other hand, the measured term values are more accurately known than the dissociation energies, we have adopted another convention in which the zero of energy is defined for all isotope combinations at the minimum of the $X\,^1\Sigma^+$ ground state. Any other definition simply requires an additional constant in the adiabatic corrections reported in this paper. For our definition of the energies, the C_1 coefficient of the adiabatic correction due to the lithium isotopes vanishes for the $X\,^1\Sigma^+$ state.

Given, however, this most precise definition of the zero of energy as the minimum of the $X\,^1\Sigma^+$ ground state, it is still useful to list the D_0 and D_e values which may be derived therefrom in Table IX. It should be clearly noted that the values listed for ^{7}LiH and ^{7}LiD are based on accurate experimental observation,[53,54] while the values for ^{6}LiH and ^{6}LiD are based on the assumption that the D_e values are the same as for the corresponding ^{7}Li molecule to within $\pm 0.1\ cm^{-1}$. This assumption appears justified on the basis of the close agreement between the T_e values of ^{6}Li and ^{7}Li molecules in Table VIII. It should also be noted that the $A\,^1\Sigma^+$ state dissociation limit of $\text{Li}(2p\,^2P_{1/2}) + \text{H}$ is 14903.66 cm^{-1} above the $X\,^1\Sigma^+$ state dissociation limit. Note in particular, that we obtain in this way not merely the electronic isotrope shift (2.75 cm^{-1} in Table VIII), but (less accurately) the individual D_e shifts in the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states. In both cases, the hydride potential well is shallower, by $5.7 \pm 0.9\ cm^{-1}$ in the $X\,^1\Sigma^+$ state and by $2.9 \pm 0.9\ cm^{-1}$ in the $A\,^1\Sigma^+$ state. The $X\,^1\Sigma^+$ value of 5.7 cm^{-1} is in reasonable agreement with the theoretical value of 4.7 cm^{-1} reported by Bardo *et al.*[28] Note that 3/4 of the theoretical result arises from electron correlation.[27,28]

In Figs. 3 and 4, we have shown the adiabatic corrections due to the hydrogen isotopes for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ state, respectively. Figure 5 shows the adiabatic correction for the $A\,^1\Sigma^+$ state due to the lithium isotope. The dashed lines in Fig. 5 give the standard error of $\Delta U_{Ad}^{1,1}$ based on the Dunham-type coefficients which is certainly smaller than the real standard error based on measured term values. All the curves are drawn with the energy convention mentioned. For the $A\,^1\Sigma^+$ state in Figs. 4 and 5, we have indicated a second abscissa

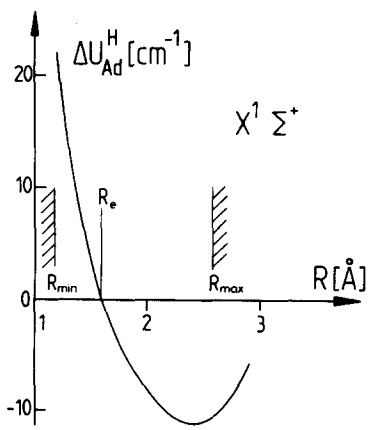


FIG. 3. The adiabatic correction of the $X\,^1\Sigma^+$ state due to the hydrogen isotopes as a function of the internuclear distance R.

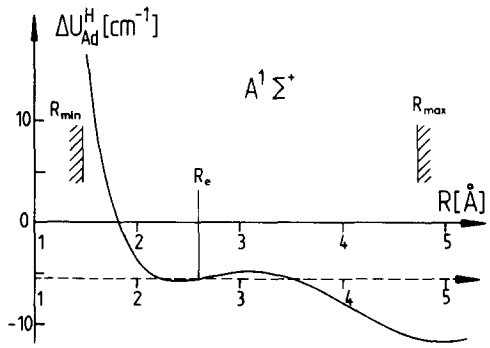


FIG. 4. The adiabatic correction of the $A\,^1\Sigma^+$ state due to the hydrogen isotopes as a function of the internuclear distance R.

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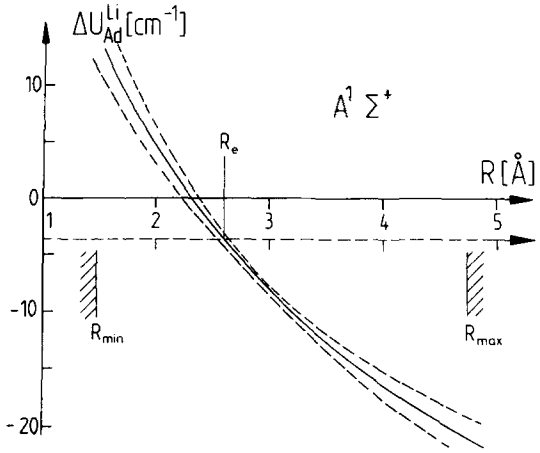


FIG. 5. The adiabatic correction of the $A\,^1\Sigma^+$ state due to the lithium isotopes as a function of the internuclear distance R. The dashed lines indicate the standard error based on the Dunham-type coefficients.

by means of a dashed line. It defines the adiabatic correction of the $A\,^1\Sigma^+$ state if one prefers a definition in which all the isotope combinations have identical energies at the minimum of the $A\,^1\Sigma^+$ state. In this case, one has to lower simultaneously the abscissa in Fig. 3 by the same amount of 5.5 cm^{-1} in order to be consistent. The shift of 5.5 cm^{-1} corresponds to the electronic isotope shift $\Delta T_e^H = 2.75$ cm^{-1} mentioned above because $2.75 = 5.5\ (M_H^{-1} - M_D^{-1})$. The shift of 3.6 cm^{-1} for the lithium isotopes corresponds to an electronic isotope shift $\Delta T_e^{Li} = 0.08$ cm^{-1} because $0.08 = 3.6\ [M^{-1}(^6\mathrm{Li}) = M^{-1}(^7\mathrm{Li})]$. Although the latter shift is not statistically significant, based on the standard error of the A_{00}^* constant in Table II, we have included it in Fig. 5 in order to make all the curves in Figs. 3, 4, and 5 consistent with the Dunham-type coefficients used in the variational method.

In Figs. 3, 4, and 5 we have also indicated the range of internuclear distances $R_{min} < R < R_{max}$ which is defined by the measurements. We also marked the position of the potential minimum R_e. For an energy convention which ties the potential energy curves of the different isotope combinations together at the potential minimum, the adiabatic correction goes, of course, through zero for $R = R_e$.

It is also interesting to note that in the case of ΔU_{Ad}^H in Figs. 3 and 4 the adiabatic correction around the potential minimum at $R = R_e$ has a very small positive derivative dU/dR for the $A\,^1\Sigma^+$ state and a negative one for the $X\,^1\Sigma^+$ state. This implies that for the $A\,^1\Sigma^+$ state the internuclear distance of the potential minimum increases slightly replacing the hydrogen atom by the deuterium atom, whereas for the $X\,^1\Sigma^+$ state the internuclear distance decreases. Similarly, we have a negative derivative at $R = R_e$ in Fig. 5 and the internuclear distance of the $A\,^1\Sigma^+$ state decreases replacing the ^{6}Li isotope by the ^{7}Li isotope. If one calculates the internuclear distance at the potential minimum from the $\bar{A}_{01}$ constant given in Table V according to

$$R_e = \sqrt{\frac{\hbar}{4\pi c \mu B_e}}\ , \tag{22}$$

one obtains the same result except for the $A\,^1\Sigma^+$ state, where the internuclear distance would also decrease replacing the hydrogen atom by the deuterium atom. This reflects again the anomalous shape of the $A\,^1\Sigma^+$ state around the potential minimum, where a B_e constant determined from a least squares fit gives a satisfactory description of the term values but not of the inertial properties of the molecule.

For the $X\,^1\Sigma^+$ state, the adiabatic correction looks qualitatively similar to what Kołos and Wolniewicz have calculated for the ground state of the H_2 molecule. One obtains the strongest influence of the adiabatic correction for the inner turning points of the potential and the adiabatic correction decreases gradually with increasing internuclear distance.

TABLE X. The IPA potential energy curve of the ^{6}LiH $A\,^1\Sigma^+$ state. $R_e = 2.59647$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	132.73	2.223609	3.011013	2.91197	132.83	2.90725
1	417.01	2.022040	3.286558	2.95009	417.00	2.95103
2	733.75	1.905558	3.472913	2.96429	733.71	2.96478
3	1073.51	1.821846	3.624418	2.96857	1073.55	2.96810
4	1430.59	1.756250	3.757337	2.96495	1430.63	2.96490
5	1800.80	1.702177	3.879158	2.95499	1800.80	2.95509
6	2180.82	1.656090	3.994044	2.93890	2180.81	2.93882
7	2567.94	1.616072	4.104534	2.91722	2567.93	2.91696
8	2959.82	1.580996	4.212311	2.89056	2959.82	2.89041
9	3354.47	1.549947	4.318622	2.85923	3354.45	2.85955
10	3750.09	1.522035	4.424467	2.82337	3750.06	2.82420
11	4145.08	1.496516	4.530657	2.78311	4145.05	2.78411
12	4537.95	1.472995	4.637859	2.73868	4537.93	2.73938
13	4927.31	1.451248	4.746703	2.69023	4927.31	2.69056
14	5311.84	1.431430	4.857934	2.63762	5311.83	2.63816
15	5690.18	1.413427	4.972440	2.58027	5690.17	2.58207
16	6060.93	1.396900	5.091220	2.51740	6060.85	2.52218

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TABLE XI. The IPA potential energy curve of the ^{7}LiH $A\,^1\Sigma^+$ state. $R_e = 2.596\,28$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	131.26	2.225 177	3.008 724	2.852 63	131.42	2.847 49
1	412.28	2.024 197	3.283 068	2.889 60	412.29	2.890 40
2	725.25	1.907 990	3.468 537	2.903 55	725.21	2.904 03
3	1060.94	1.824 439	3.619 243	2.907 92	1060.97	2.907 47
4	1413.74	1.758 948	3.751 378	2.904 68	1413.77	2.904 62
5	1779.57	1.704 955	3.872 396	2.895 32	1779.57	2.895 45
6	2155.19	1.658 925	3.986 441	2.880 06	2155.17	2.880 04
7	2537.91	1.618 933	4.096 045	2.859 41	2537.90	2.859 20
8	2925.47	1.583 854	4.202 884	2.833 94	2925.46	2.833 82
9	3315.89	1.552 796	4.308 190	2.803 98	3315.87	2.804 28
10	3707.44	1.524 891	4.412 951	2.769 66	3707.40	2.770 48
11	4098.53	1.499 396	4.517 974	2.731 09	4098.48	2.732 14
12	4487.70	1.475 850	4.623 914	2.688 50	4487.67	2.689 31
13	4873.61	1.454 120	4.731 379	2.642 06	4873.60	2.642 47
14	5254.97	1.434 238	4.841 070	2.591 68	5254.95	2.592 16
15	5630.47	1.416 158	4.953 833	2.536 86	5630.44	2.538 36

For the $A\,^1\Sigma^+$ state, one obtains a more peculiar shape of the adiabatic correction. In particular, one obtains a shoulder which looks like a broad intermediate maximum superimposed on a decaying slope. An intermediate maximum has been obtained by Kołos and Wolniewicz for the double minimum potential of the E, F state of the H_2 molecule (see Fig. 1 of Ref. 11). In their case, the rather dramatic maximum occurs at an internuclear distance, where the electronic wave function changes drastically its character. In our case, something similar, although by far less dramatic, may occur. It has to be less dramatic than in case of the H_2, E, F state because otherwise the adiabatic approximation would no longer be valid.

It should be noted that the magnitude and sign of the adiabatic corrections reported here is the same as noted in the much cruder treatment of the $X\,^1\Sigma^+$ state only by Li and Stwalley.[9] However, the uncertainty is now precisely defined and greatly reduced in magnitude so that structure in the $A\,^1\Sigma^+$ adiabatic corrections in particular can now be discerned.

It is interesting to note that the maximum of the adiabatic correction occurs at an internuclear separation of $R \simeq 3.2$ Å, which is very close to where *ab initio* calculations[7,55] show that the electric dipole moment of the $A\,^1\Sigma^+$ state goes through zero and changes sign, and to where (3.9 Å) the ionic–covalent crossing point occurs.[50,51,56]

In Tables X–XVII, we summarize the potential energy curves of the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ states for the four different isotope combinations. In some cases we have given a few more energy levels $E_{v,J=0}$ than expected from the

TABLE XII. The IPA potential energy curve of the ^{6}LiD $A\,^1\Sigma^+$ state. $R_e = 2.597\,10$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	98.22	2.267 716	2.957 256	1.664 92	97.81	1.664 15
1	306.59	2.080 768	3.200 829	1.683 69	306.55	1.684 42
2	535.94	1.971 111	3.364 486	1.692 70	535.90	1.693 25
3	781.03	1.891 575	3.496 148	1.697 53	781.04	1.697 30
4	1038.63	1.828 701	3.610 165	1.699 15	1038.64	1.698 80
5	1306.31	1.776 633	3.713 068	1.698 39	1306.32	1.698 18
6	1582.20	1.732 138	3.808 515	1.695 36	1582.20	1.695 33
7	1864.74	1.693 215	3.898 792	1.690 22	1864.73	1.690 23
8	2152.59	1.658 602	3.985 424	1.683 13	2152.59	1.683 03
9	2444.58	1.627 512	4.069 467	1.674 25	2444.59	1.674 01
10	2739.68	1.599 430	4.151 696	1.663 72	2739.70	1.663 40
11	3036.98	1.573 946	4.232 715	1.651 56	3037.00	1.651 34
12	3335.65	1.550 663	4.313 021	1.637 81	3335.65	1.637 83
13	3634.93	1.529 191	4.393 042	1.622 51	3634.92	1.622 79
14	3934.13	1.509 188	4.473 143	1.605 76	3934.11	1.606 17
15	4232.56	1.490 403	4.553 633	1.587 66	4232.55	1.587 97
16	4529.60	1.472 706	4.634 782	1.568 33	4529.61	1.568 34
17	4824.64	1.456 071	4.716 860	1.547 81	4824.66	1.547 40
18	5117.10	1.440 522	4.800 177	1.526 02	5117.13	1.525 11

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TABLE XIII. The IPA potential energy curve of the ^{7}LiD $A\,^1\Sigma^+$ state. $R_e = 2.596\,90$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	96.33	2.270 283	2.953 815	1.605 46	95.98	1.604 41
1	300.56	2.084 369	3.195 477	1.623 28	300.55	1.623 88
2	525.17	1.975 197	3.357 804	1.631 95	525.14	1.632 48
3	765.11	1.895 965	3.488 324	1.636 68	765.12	1.636 47
4	1017.27	1.833 296	3.601 275	1.638 39	1017.30	1.638 05
5	1279.33	1.781 373	3.703 135	1.637 85	1279.34	1.637 64
6	1549.46	1.736 993	3.797 529	1.635 21	1549.45	1.635 18
7	1826.16	1.698 167	3.886 725	1.630 58	1826.15	1.630 61
8	2108.15	1.663 627	3.972 239	1.624 11	2108.14	1.624 07
9	2394.29	1.632 579	4.055 121	1.615 98	2394.29	1.615 78
10	2683.59	1.604 504	4.136 140	1.606 29	2683.60	1.606 00
11	2975.16	1.579 007	4.215 889	1.595 07	2975.18	1.594 85
12	3268.23	1.555 711	4.294 859	1.582 37	3268.24	1.582 36
13	3562.04	1.534 244	4.373 466	1.568 22	3562.03	1.568 47
14	3855.93	1.514 267	4.452 070	1.552 68	3855.91	1.553 11
15	4149.25	1.495 515	4.530 974	1.535 87	4149.24	1.536 28
16	4441.40	1.477 836	4.610 444	1.517 91	4441.40	1.518 07
17	4731.78	1.461 180	4.690 728	1.498 84	4731.80	1.498 61
18	5019.86	1.445 564	4.772 105	1.478 64	5019.88	1.477 95

TABLE XIV. The IPA potential energy curve of the ^{6}LiH $X\,^1\Sigma^+$ state. $R_e = 1.595\,56$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	705.17	1.444 905	1.779 130	7.560 09	705.16	7.560 31
1	2 078.57	1.352 041	1.938 836	7.340 97	2 078.56	7.341 11
2	3 406.17	1.295 449	2.063 743	7.125 79	3 406.18	7.125 73
3	4 688.90	1.253 455	2.175 610	6.914 31	4 688.93	6.914 07
4	5 927.63	1.219 839	2.280 754	6.706 21	5 927.66	6.705 91
5	7 123.17	1.191 794	2.382 102	6.501 12	7 123.19	6.500 87
6	8 276.18	1.167 778	2.481 347	6.298 56	8 276.19	6.298 46
7	9 387.22	1.146 845	2.579 627	6.097 96	9 387.21	6.098 05
8	10 456.64	1.128 368	2.677 812	5.898 61	10 456.63	5.898 85
9	11 484.62	1.111 907	2.776 650	5.699 65	11 484.62	5.699 95
10	12 471.09	1.097 133	2.876 858	5.500 06	12 471.10	5.500 31
11	13 415.71	1.083 791	2.979 161	5.298 73	13 415.75	5.298 73
12	14 317.91	1.071 681	3.084 336	5.094 49	14 317.91	5.093 90
13	15 176.84	1.060 655	3.193 273	4.885 95	15 176.61	4.884 36

TABLE XV. The IPA potential energy curve of the ^{7}LiH $X\,^1\Sigma^+$ state. $R_e = 1.595\,56$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	697.96	1.445 610	1.778 085	7.406 44	697.96	7.406 66
1	2 057.71	1.353 096	1.936 737	7.193 97	2 057.70	7.194 12
2	3 372.57	1.296 683	2.060 732	6.985 29	3 372.58	6.985 25
3	4 643.45	1.254 799	2.171 721	6.780 16	4 643.48	6.779 94
4	5 871.22	1.221 257	2.275 984	6.578 29	5 871.25	6.578 00
5	7 506.64	1.193 263	2.376 434	6.379 33	7 056.66	6.379 09
6	8 200.37	1.169 280	2.474 750	6.182 82	8 200.38	6.182 72
7	9 302.95	1.148 367	2.572 056	5.988 23	9 302.94	5.988 32
8	10 364.76	1.129 898	2.669 208	5.794 91	10 364.74	5.795 14
9	11 385.96	1.113 437	2.766 940	5.602 03	11 385.94	5.602 33
10	12 366.52	1.098 656	2.865 943	5.408 65	12 366.52	5.408 91
11	13 306.13	1.085 302	2.966 921	5.213 72	13 306.16	5.213 77
12	14 204.27	1.073 176	3.070 618	5.016 14	14 204.27	5.015 66
13	15 060.16	1.062 126	3.177 879	4.814 63	15 059.96	4.813 22

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TABLE XVI. The IPA potential energy curve of the ^{6}LiD $X\,^1\Sigma^+$ state. $R_e = 1.595\,27$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	534.33	1.462 635	1.752 748	4.342 03	534.33	4.342 04
1	1 581.82	1.379 002	1.886 650	4.246 82	1 581.82	4.246 79
2	2 602.91	1.327 204	1.989 549	4.152 90	2 602.93	4.152 79
3	3 598.03	1.288 290	2.080 431	4.060 21	3 598.06	4.060 04
4	4 567.57	1.256 804	2.164 790	3.968 69	4 567.59	3.968 49
5	5 511.90	1.230 274	2.245 140	3.878 26	5 511.90	3.878 10
6	6 431.36	1.207 334	2.322 897	3.788 83	6 431.34	3.788 79
7	7 326.27	1.187 142	2.398 962	3.700 30	7 326.23	3.700 43
8	8 196.90	1.169 135	2.473 968	3.612 54	8 196.86	3.612 91

TABLE XVII. The IPA potential energy curve of the ^{7}LiD $X\,^1\Sigma^+$ state. $R_e = 1.595\,27$ Å.

					Least squares fit	
v	G_v (cm^{-1})	R_- (Å)	R_+ (Å)	B_v (cm^{-1})	G_v (cm^{-1})	B_v (cm^{-1})
0	524.74	1.463 740	1.751 192	4.187 49	524.73	4.187 50
1	1 553.78	1.380 690	1.883 589	4.097 32	1 553.78	4.097 30
2	2 557.37	1.329 204	1.985 223	4.008 35	2 557.38	4.008 25
3	3 535.89	1.290 495	2.074 915	3.920 53	3 535.91	3.920 37
4	4 489.72	1.259 154	2.158 108	3.833 80	4 489.74	3.833 62
5	5 419.22	1.232 731	2.237 294	3.748 09	5 419.22	3.747 94
6	6 324.71	1.209 871	2.313 872	3.663 32	6 324.69	3.663 27
7	7 206.49	1.189 738	2.388 734	3.579 40	7 206.46	3.579 50
8	8 064.83	1.171 774	2.462 504	3.496 21	8 064.79	3.496 53

data fields in Fig. 1. If one exploits, however, the fact that for the $X\,^1\Sigma^+$ state the potential energy curves are identical for the lithium hydrides ^{6}LiH and ^{7}LiH and similarly for the lithium deuterides ^{6}LiD and ^{7}LiD, then these additional term values are defined by the measurements containing the other lithium isotope. Future work on the lithium tritides is planned.

Finally, we have investigated the origin of the rather dramatic improvement of the IPA potentials with respect to the RKR potentials. First of all, we had to exclude the possibility that the very large rms error σ_E(RKR) of the RKR potential is simply a consequence of inadequate G_v and B_v values as obtained from the least squares fit through the $\bar{A}_{ik}$ constants. A situation of this kind occurred, for example, in case of the $X\,^1\Sigma_g^+$ state of the Mg_2 molecule.[32] However, it turns out that this is not the case for the $A\,^1\Sigma^+$ state of the lithium hydrides. For this purpose, we calculated G_v and B_v values from the IPA potentials. The G_v values are given by the energy eigenvalues listed in the second column of Tables X–XVII

$$G_v = E_{v,J=0} \tag{23}$$

and the B_v values are given by the expectation values of $1/R^2$ of the rotationless molecule

$$B_v = \frac{\hbar}{4\pi\mu c} \langle \psi_{v,J=0} | R^{-2} | \psi_{v,J=0} \rangle , \tag{24}$$

which are listed in the fifth column of Tables X–XVII. Using these G_v and B_v values, we constructed RKR potentials which revealed an rms error σ_E(RKR) which was just as large as the original rms errors summarized in Table VII.

Comparing then the rms error σ_E(RKR) for the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ state, it appears that the very large rms error for the $A\,^1\Sigma^+$ state is not an inherent deficiency of the WKB approximation, but rather a consequence of the anomalous shape of the potential minimum. It is certainly the origin for the difference of the G_v and B_v values of the IPA potentials which is very much larger for the $A\,^1\Sigma^+$ state than it is for the $X\,^1\Sigma^+$ state (see Tables X–XVII).

Finally, it should be noted that values of R_e are given in the caption of Tables X–XVII. For these, the IPA method does not give really accurate values compared to the accuracy of the turning points R_+ and R_-. This is because, although the minimum energy in the potentials are accurately defined, the slope dV/dR at the minimum is zero, which amplifies the uncertainty in the R_e value.

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